Colimits of Enriched Categories

Andy Sukowski-Bang — supervised by Timo Essig

arxiv.org/abs/2607.07780

Question

What is the colimit of a diagram $\mathcal{I} \to \mathcal{V}$-Cat?

Question

What is the colimit of a diagram $\mathcal{I} \to \mathcal{V}\text{-Cat}$?

category of $\mathcal{V}$-enriched categories

Question

What is the colimit of a diagram $\mathcal{I} \to \mathcal{V}\text{-Cat}$?

category of $\mathcal{V}$-enriched categories
(hom-objects in $\mathcal{V}$ instead of hom-sets)

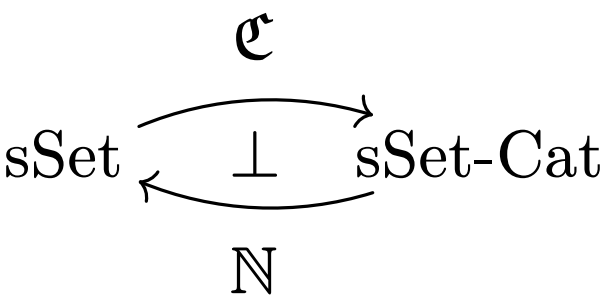


6

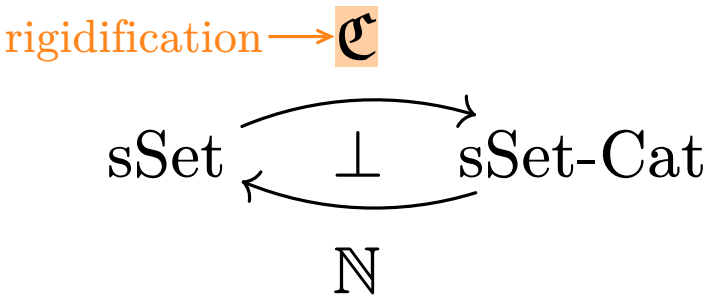
How to compute (co)limits of enriched categories?

Asked 11 years ago Modified 1 month ago Viewed 829 times

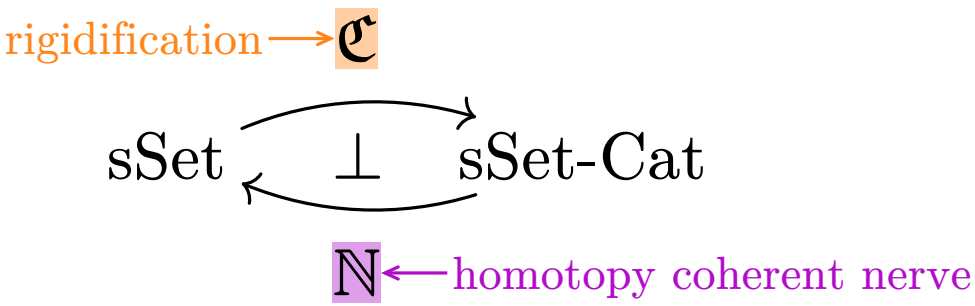
Rigidification via Necklaces



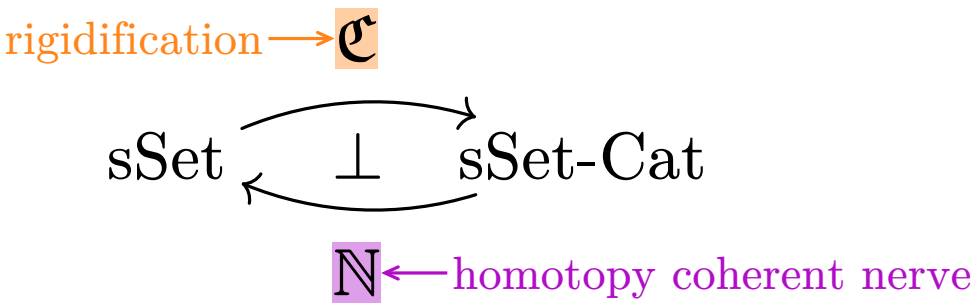
Rigidification via Necklaces



Rigidification via Necklaces

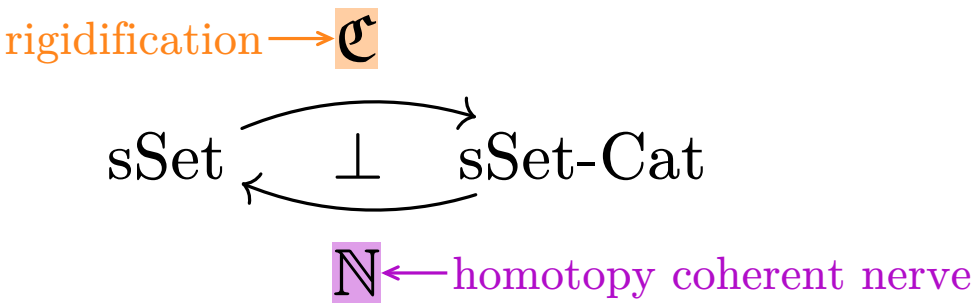


Rigidification via Necklaces



$$\mathfrak{C}[X] := \operatorname{colim}_{\Delta^n \to X} \mathfrak{C}[\Delta^n]$$

Rigidification via Necklaces



$$\mathfrak{C}[X] := \operatorname{colim}_{\Delta^n \to X} \mathfrak{C}[\Delta^n]$$

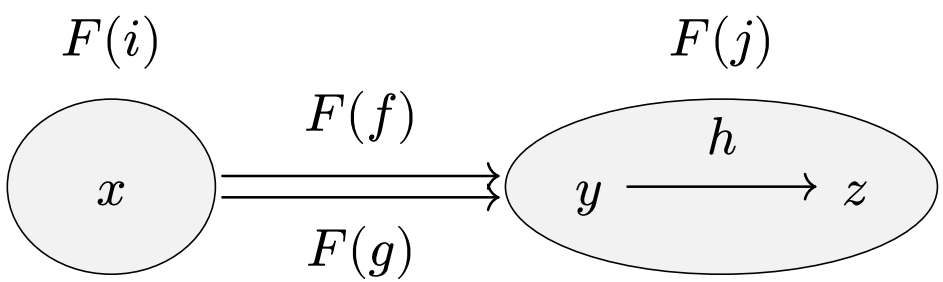
Dugger-Spivak, necklaces in *Rigidification of quasi-categories*

Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.

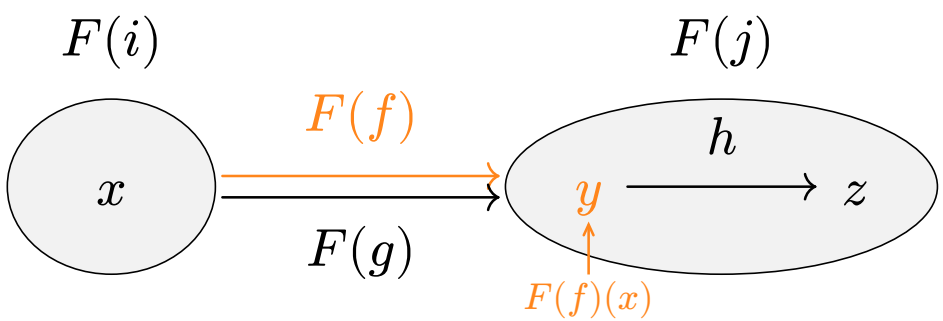
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



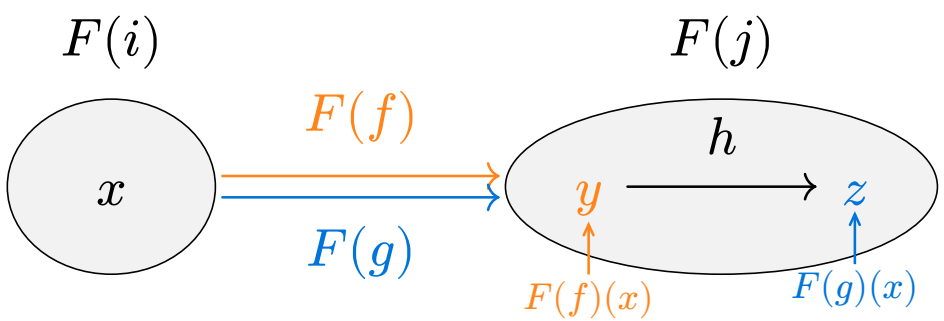
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



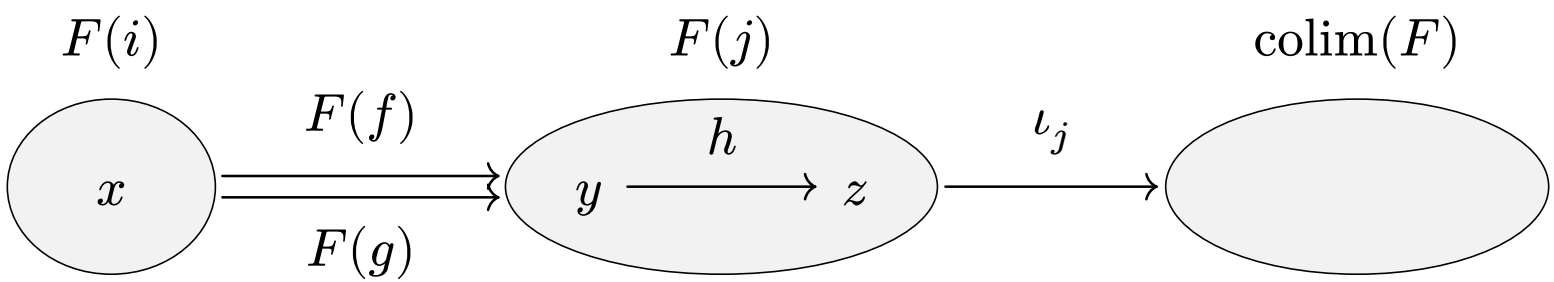
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



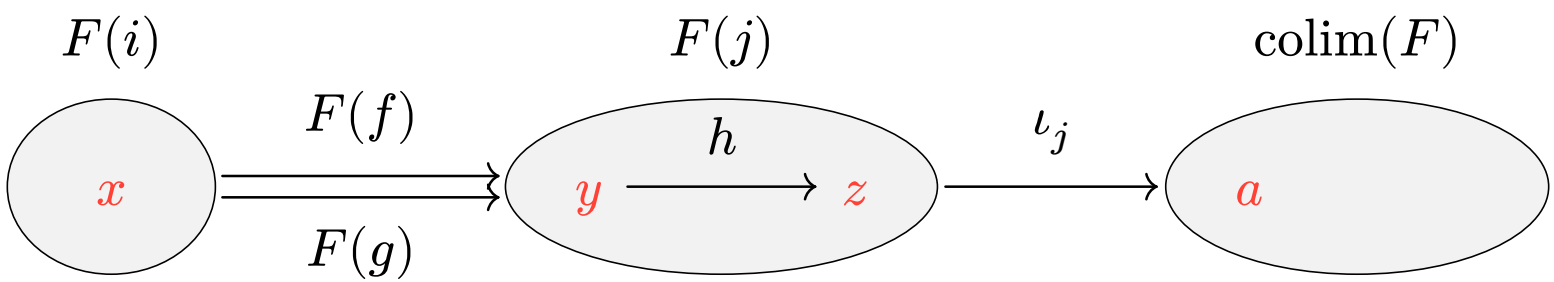
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



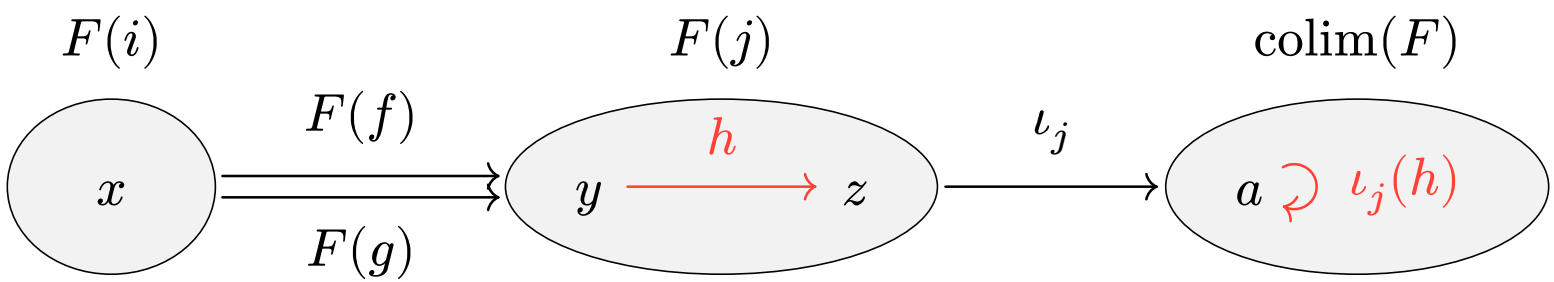
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



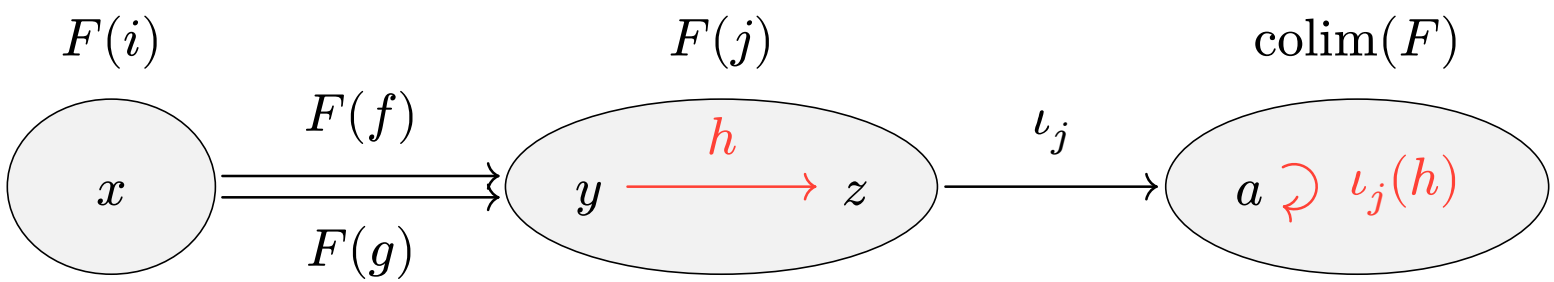
Coequalizer in Cat

Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



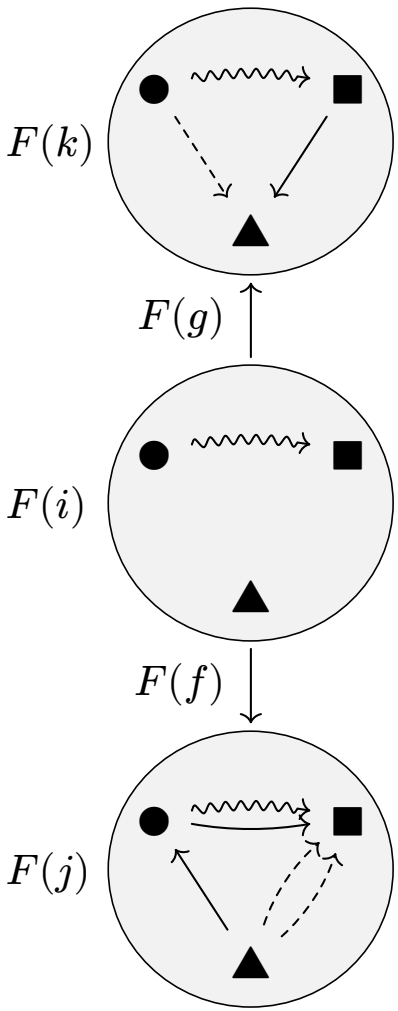
Coequalizer in Cat

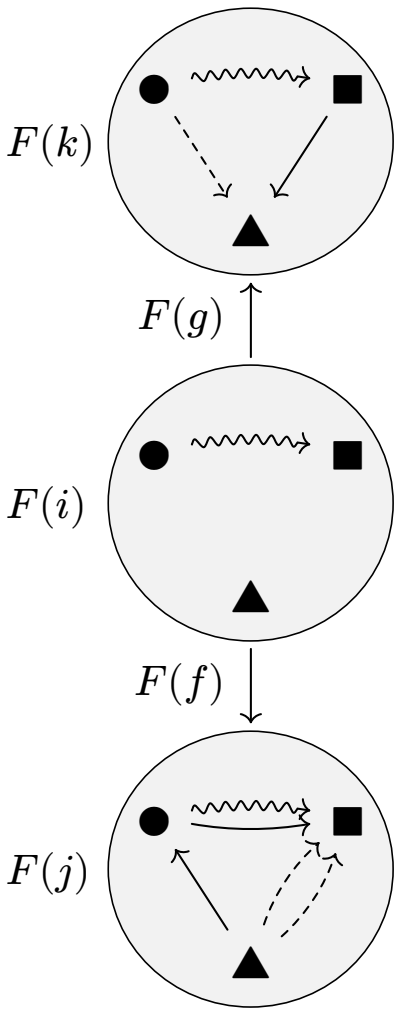
Let $\mathcal{I}$ consist of $f, g\colon i \to j$ and consider $F\colon \mathcal{I} \to \mathrm{Cat}$.



Infinitely many non-trivial compositions $\iota_j(h) \circ \cdots \circ \iota_j(h) \ldots$

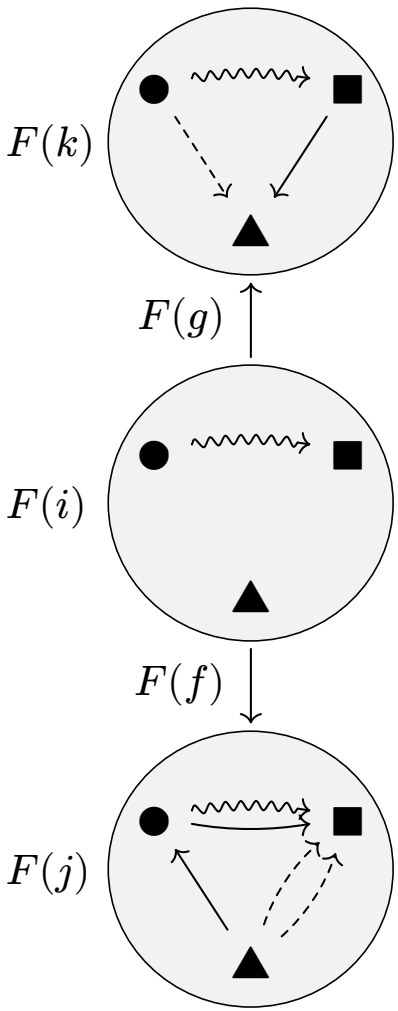
Colimits in Cat





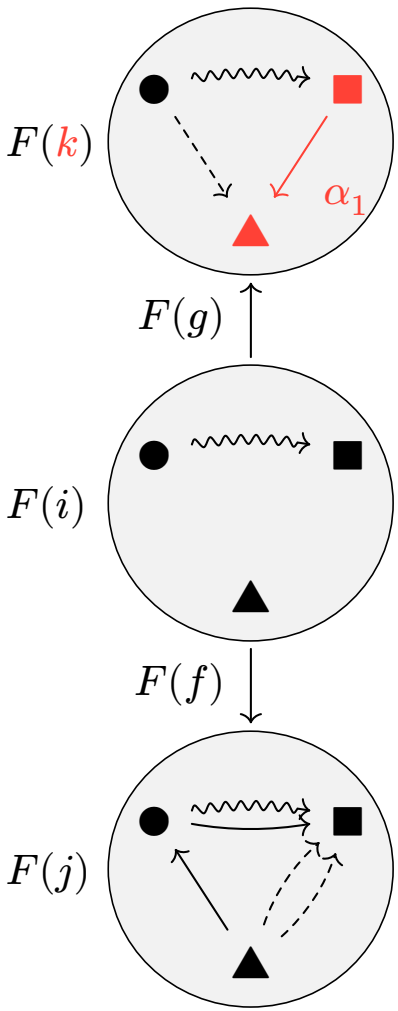
Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$



Colimits in Cat

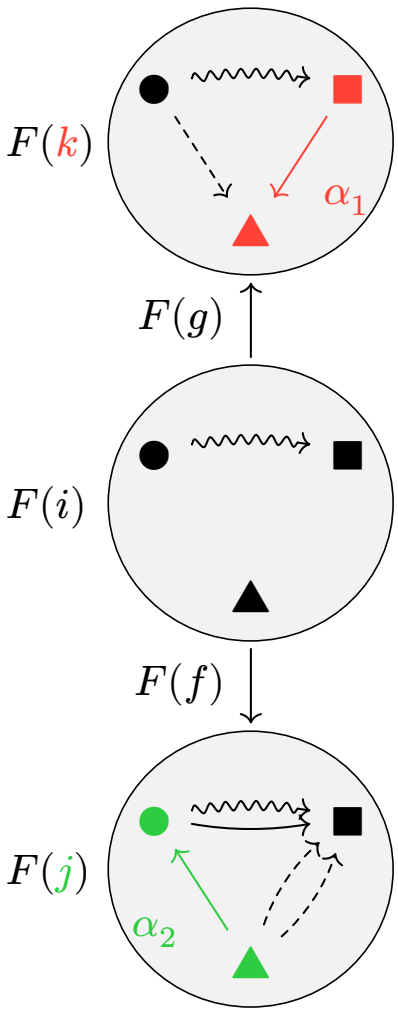
- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.



Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

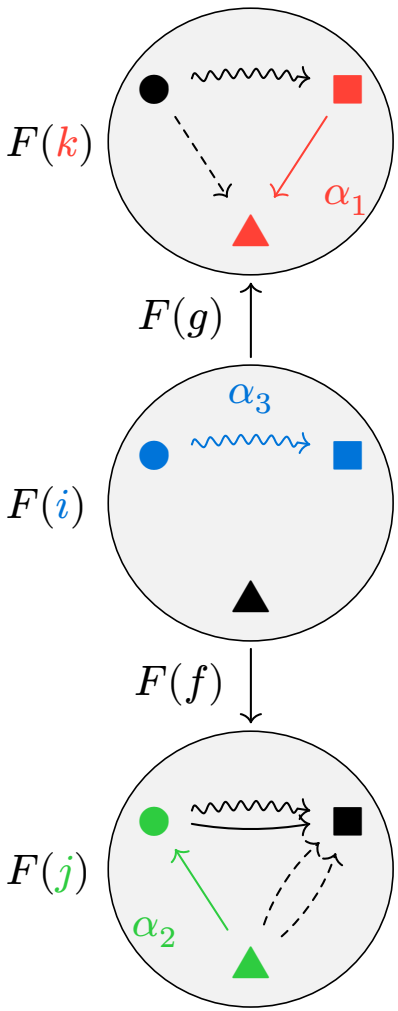
  Example. $[(k, \alpha_1), ...]$



Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

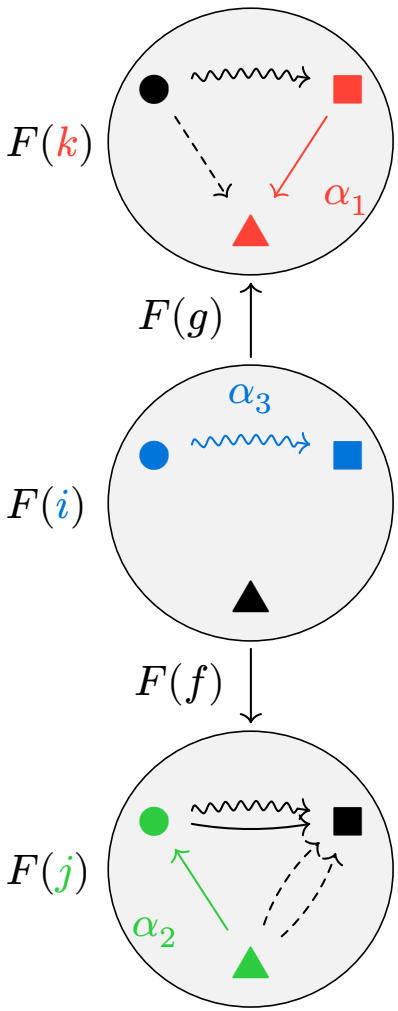
  Example. $[(k, \alpha_1), (j, \alpha_2), ...]$



Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

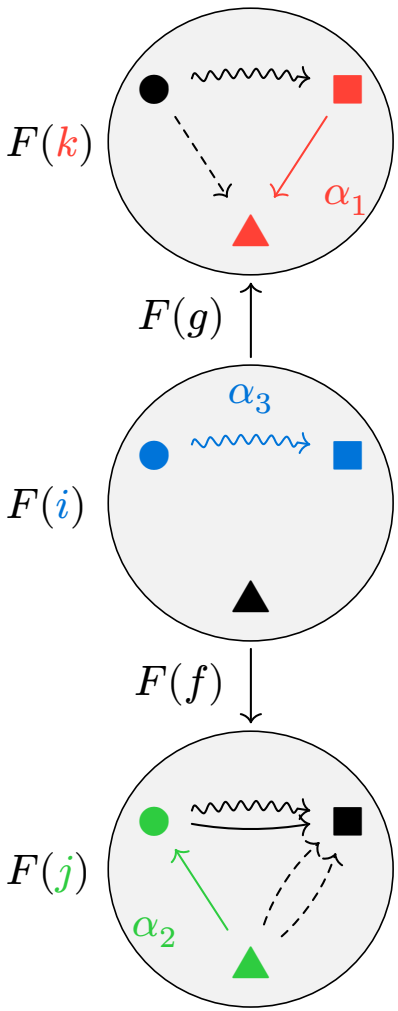


Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,

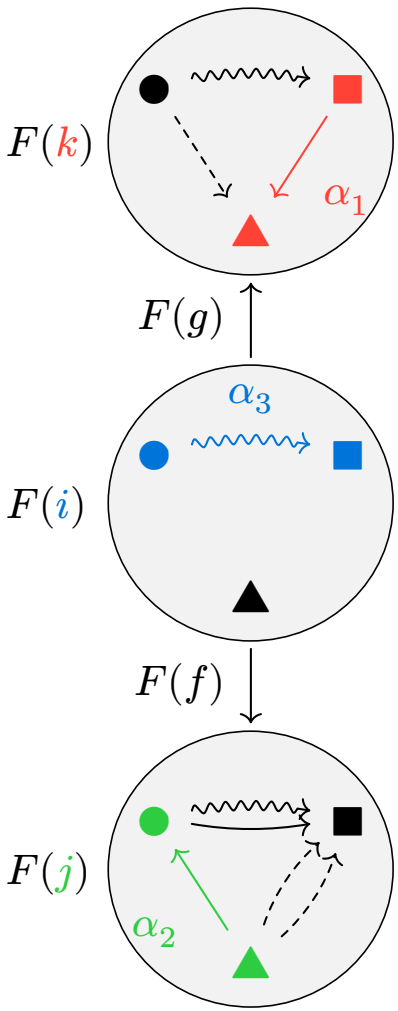


Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,
  - $\mathrm{dom}(\alpha_1) \in a$ and $\mathrm{cod}(\alpha_n) \in b$,

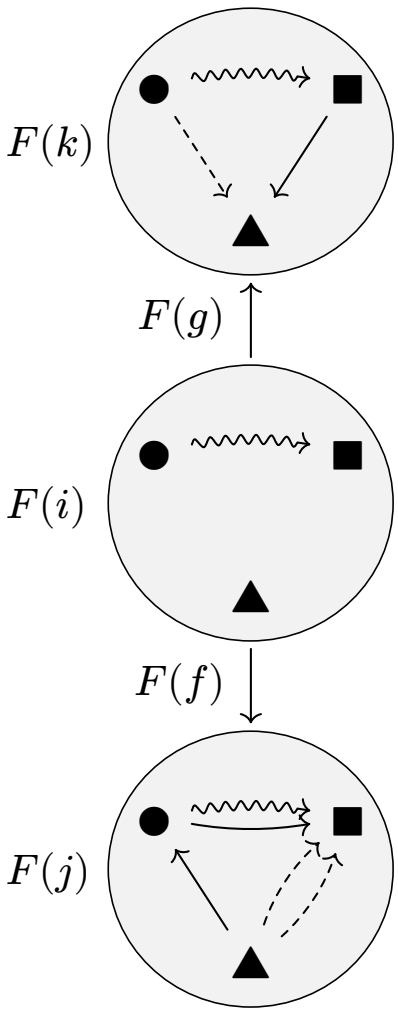


Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,
  - $\mathrm{dom}(\alpha_1) \in a$ and $\mathrm{cod}(\alpha_n) \in b$,
  - $\mathrm{cod}(\alpha_{k-1}) \sim \mathrm{dom}(\alpha_k)$.

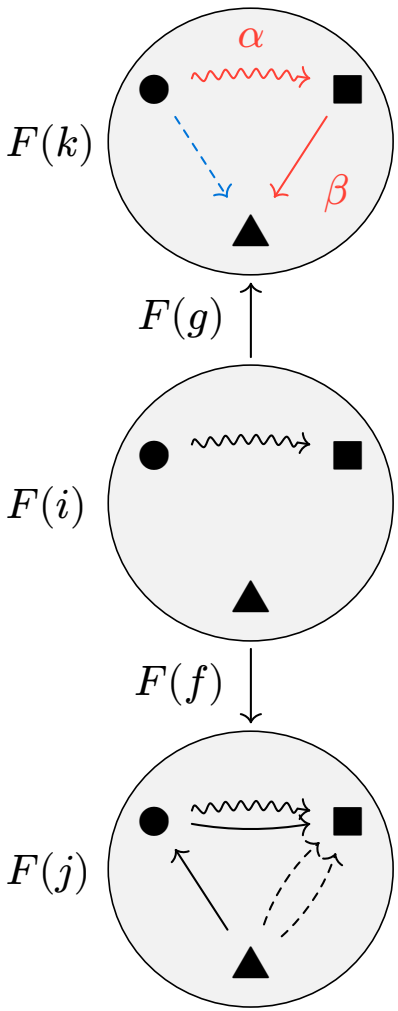


Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,
  - $\mathrm{dom}(\alpha_1) \in a$ and $\mathrm{cod}(\alpha_n) \in b$,
  - $\mathrm{cod}(\alpha_{k-1}) \sim \mathrm{dom}(\alpha_k)$.
- The equivalence relation is generated by…

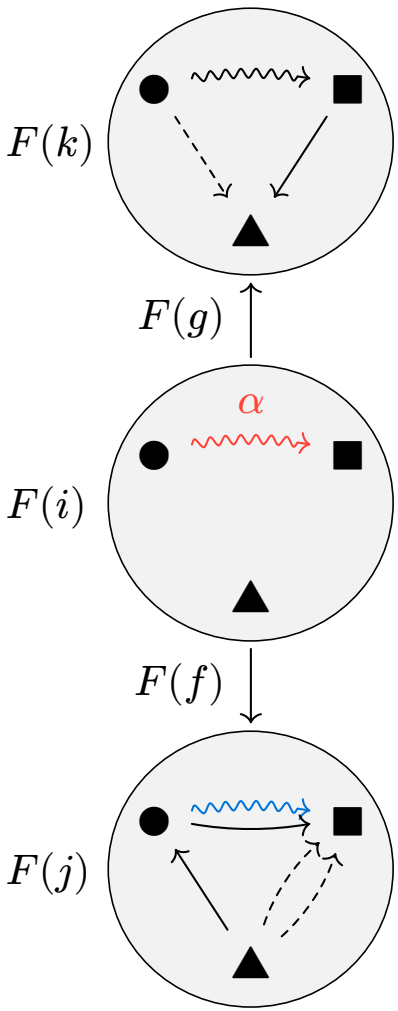


Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,
  - $\mathrm{dom}(\alpha_1) \in a$ and $\mathrm{cod}(\alpha_n) \in b$,
  - $\mathrm{cod}(\alpha_{k-1}) \sim \mathrm{dom}(\alpha_k)$.
- The equivalence relation is generated by
  - $[..., (k, \alpha), (k, \beta), ...] \sim [..., (k, \beta \circ \alpha), ...]$,



Colimits in Cat

- $\mathrm{Ob}(\operatorname{colim} F) \cong \operatorname{colim}(\mathrm{Ob} \circ F)$
- Morphisms in $(\operatorname{colim} F)(a, b)$ are equivalence classes of strings $[(i_1, \alpha_1), ..., (i_n, \alpha_n)]$.

  Example. $[(k, \alpha_1), (j, \alpha_2), (i, \alpha_3)]$

  - $\alpha_k \in \operatorname{mor} F(i_k)$,
  - $\mathrm{dom}(\alpha_1) \in a$ and $\mathrm{cod}(\alpha_n) \in b$,
  - $\mathrm{cod}(\alpha_{k-1}) \sim \mathrm{dom}(\alpha_k)$.
- The equivalence relation is generated by
  - $[..., (k, \alpha), (k, \beta), ...] \sim [..., (k, \beta \circ \alpha), ...]$,
  - $[..., (i, \alpha), ...] \sim [..., (j, (Ff)(\alpha)), ...]$.

Colimits in $\mathcal{V}$-Cat

Colimits in $\mathcal{V}$-Cat

$$(\operatorname{colim} F)(a,b) \cong \operatorname*{colim}_{\mathcal{S}(a,b)} (\mathrm{ev})$$

Colimits in $\mathcal{V}$-Cat

$$(\operatorname{colim} F)(a,b) \cong \operatorname*{colim}_{\mathcal{S}(a,b)} (\mathrm{ev})$$

- category of strings $\mathcal{S}(a,b)$ has objects

$$a \xrightarrow{(i_1,x_1,y_1)} \bullet \xrightarrow{(i_2,x_2,y_2)} \cdots \xrightarrow{(i_n,x_n,y_n)} b. \tag{1}$$

Colimits in $\mathcal{V}$-Cat

$$(\operatorname{colim} F)(a,b) \cong \operatorname*{colim}_{\mathcal{S}(a,b)} (\mathrm{ev})$$

- category of strings $\mathcal{S}(a,b)$ has objects

$$a \xrightarrow{(i_1,x_1,y_1)} \bullet \xrightarrow{(i_2,x_2,y_2)} \cdots \xrightarrow{(i_n,x_n,y_n)} b. \qquad (1)$$

- string evaluation ev: $\mathcal{S}(a,b) \to \mathcal{V}$ sends the string from (1) to

$$F(i_n)(x_n,y_n) \otimes \cdots \otimes F(i_1)(x_1,y_1).$$

Composition

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \cong \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \xrightarrow{?} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Composition

fubini for colimits

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \cong \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)} (\mathrm{ev} \boxtimes \mathrm{ev}) \xrightarrow{?} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Composition

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \overset{\text{fubini for colimits}}{\cong} \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \overset{?}{\longrightarrow} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Compare colimits of varying shapes via Grothendieck construction

$$\text{colim: } \int \mathrm{Fun}(-, \mathcal{V}) \to \mathcal{V}.$$

Composition

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \overset{\text{fubini for colimits}}{\cong} \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \overset{?}{\longrightarrow} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Compare colimits of varying shapes via Grothendieck construction

$$\text{strong monoidal} \longrightarrow \text{colim:} \int \mathrm{Fun}(-, \mathcal{V}) \to \mathcal{V}.$$

Composition

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \overset{\text{fubini for colimits}}{\cong} \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \overset{?}{\longrightarrow} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Compare colimits of varying shapes via Grothendieck construction

$$\text{strong monoidal} \longrightarrow \text{colim: } \int \mathrm{Fun}(-, \mathcal{V}) \to \mathcal{V}.$$

- Objects $(\mathcal{I}, F) \mapsto \operatorname{colim}_{\mathcal{I}}(F)$.

Composition

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \overset{\text{fubini for colimits}}{\cong} \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \overset{?}{\longrightarrow} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Compare colimits of varying shapes via Grothendieck construction

$$\text{strong monoidal} \longrightarrow \text{colim: } \int \mathrm{Fun}(-, \mathcal{V}) \to \mathcal{V}.$$

- Objects $(\mathcal{I}, F) \mapsto \operatorname{colim}_{\mathcal{I}}(F)$.
- Morphism $\operatorname{colim}(H, \eta)$ with $H\colon \mathcal{I} \to \mathcal{J}$ and $\eta\colon F \Rightarrow G \circ H$ is induced by

$$F \overset{\eta}{\Longrightarrow} G \circ H \overset{\iota H}{\Longrightarrow} \Delta_{\mathcal{I}} \operatorname{colim}(G).$$

Composition

fubini for colimits

$$\operatorname*{colim}_{\mathcal{S}(b,c)}(\mathrm{ev}) \otimes \operatorname*{colim}_{\mathcal{S}(a,b)}(\mathrm{ev}) \cong \operatorname*{colim}_{\mathcal{S}(b,c)\times\mathcal{S}(a,b)}(\mathrm{ev} \boxtimes \mathrm{ev}) \xrightarrow{?} \operatorname*{colim}_{\mathcal{S}(a,c)}(\mathrm{ev})$$

Compare colimits of varying shapes via Grothendieck construction

strong monoidal $\longrightarrow$ colim: $\int \mathrm{Fun}(-, \mathcal{V}) \to \mathcal{V}.$

- Objects $(\mathcal{I}, F) \mapsto \operatorname{colim}_{\mathcal{I}}(F)$.
- Morphism $\operatorname{colim}(H, \eta)$ with $H\colon \mathcal{I} \to \mathcal{J}$ and $\eta\colon F \Rightarrow G \circ H$ induced by

$$F \overset{\eta}{\Longrightarrow} G \circ H \overset{\iota H}{\Longrightarrow} \Delta_{\mathcal{I}} \operatorname{colim}(G).$$

whisker cocone ι of $\operatorname{colim}(G)$

Lawvere Metric Spaces

M enriched in poset $([0, \infty], \geq)$, where...

Lawvere Metric Spaces

M enriched in poset $([0, \infty], \geq)$, where

- objects are *points*,

Lawvere Metric Spaces

M enriched in poset $([0, \infty], \geq)$, where

- objects are *points*,
- hom-objects $M(a, b)$ are *distances*,

Lawvere Metric Spaces

M enriched in poset $([0, \infty], \geq)$, where

- objects are *points*,
- hom-objects $M(a, b)$ are *distances*,
- the tensor product is addition $+$,

Lawvere Metric Spaces

M enriched in poset $([0, \infty], \geq)$, where

- objects are *points*,
- hom-objects $M(a, b)$ are *distances*,
- the tensor product is addition $+$,
- composition is the triangle identity $M(b, c) + M(a, b) \geq M(a, c)$.

Lawvere Metric Spaces

M enriched in poset $([0,\infty],\geq)$, where

- objects are *points*,
- hom-objects $M(a,b)$ are *distances*,
- the tensor product is addition $+$,
- composition is the triangle identity $M(b,c)+M(a,b)\geq M(a,c)$.

Let $F\colon \mathcal{I}\to[0,\infty]$-Cat with $i\mapsto M_i$. Then...

Lawvere Metric Spaces

M enriched in poset $([0,\infty],\geq)$, where

- objects are *points*,
- hom-objects $M(a,b)$ are *distances*,
- the tensor product is addition $+$,
- composition is the triangle identity $M(b,c)+M(a,b)\geq M(a,c)$.

Let $F\colon \mathcal{I}\to[0,\infty]$-Cat with $i\mapsto M_i$. Then

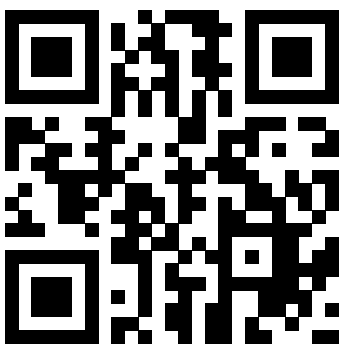
$$(\operatorname{colim} F)(a,b)=\inf\Big\{M_{i_n}(x_n,y_n)+\cdots+M_{i_1}(x_1,y_1)\Big\}.$$

arXiv Preprint



arxiv.org/abs/2607.07780

MathOverflow Answer



mathoverflow.net/a/513164